

TAKYI-AMPONSEM YEBOAH DEON

deontakyiampomsem@gmail.com | +233 592763423 | LinkedIn: Deon Takyi-Amponsem

PERSONAL STATEMENT

An innovative, aspiring engineer driven by the belief that engineering's true potential lies in developing practical solutions to real-world problems. Drawing on my skills, I applied AI-powered frameworks to classify high flow zones in subsurface rocks and went on to build a machine-learning web application that eliminated the need for manual coding. Tackling these complex projects taught me resilience, the value of hard work, and the rewards that come from perseverance, qualities I carry forward as I strive to improve, add measurable impact to every organization I am associated with, and contribute meaningfully to society at large.

PROFESSIONAL EXPERIENCE

SOH ENERGY

October 2025- Present.

National Service Personnel (Business Development Department)

- Built a manpower recruitment AI-agent form that screens and ranks CVs, automating shortlisting and emails.
- Co-developed a financial feasibility model for an urban placed fuel station, incorporating demand forecasting, cost analysis, and profitability analysis.

AI-DPRS RESEARCH LAB

September 2024-Present.

Reservoir Characterization lead

- Assisted in developing a novel and automated approach for Hydraulic flow unit (HFU) identification from core and well data using Autoencoders and unsupervised machine learning.
- Work published in the Journal of Applied Geophysics; <https://doi.org/10.1016/j.jappgeo.2025.105778>
- Applied numerical methods with conventional approaches to automate detection of high flow zones in petroleum reservoirs
- Collaborated with GNPC to use unsupervised machine learning to enhance reservoir characterization
- Developed a software that uses Machine learning for HFU and net pay delineation -RescharAI

Booster Station, Tema

November 2023- January 2024

Intern

- Assisted in tracking boosted products to various Terminals such as; Tema Oil Refinery, TOT, TTF etc.
- Performed periodic checks on pumps and valves.

Tema Tank Farm, Chase Logistics Subsidiary

September 2023- November 2023

Intern (Operations Department)

- Assisted in calibrating flow meters before offloading using proval tanks.
- Carried out periodic checks on receipt and transfer pumps, valves, pipelines and tanks.
- Helped in preparing tanks for product receipt and transfer. Products included gasoline, gas oil and light crude oil (LCO).

ACADEMIC QUALIFICATION

Bachelor of Science-Petroleum Engineering

January 2021- August 2025.

Kwame Nkrumah University of Science and Technology, Kumasi- Ghana

W.A.S.S.C.E (General Science)

September 2018 – October 2021

Mfantsipim School, Cape-Coast-Ghana

PROJECTS AND PUBLICATIONS

RescharAI

- Developed a Software for Hydraulic flow unit identification using unsupervised machine learning and net pay delineation using supervised machine learning together with residual correctors.

Publication in the Journal of Applied Geophysics.

- Automated Hydraulic Flow Unit Determination Using FZI-Z-Score Probability, K-Means Clustering with Autoencoders, and Machine Learning Classification: A Case Study of the Subei Basin, China. (<https://doi.org/10.1016/j.jappgeo.2025.105778>)

Fortel

- Co-developed an intensive machine learning web application, featuring secure, private data uploads (link: <http://fortel.onrender.com>).
- Designed entire front-end with CSS, html and JavaScript.

- *Integrating React.js and Node.js for continuous improvements.*

C7+ Compressibility (Z) Factor

- *Developed python Script to automate the rigorous calculation of the compressibility (z) of C7+ fractions.*
- *Used oil field units to standardize process.*

ACHIEVEMENTS AND AWARDS

KNUST, Bsc. Petroleum Engineering- First Class Honours

KNUST College of Engineering Excellence Award 2021/2022 academic year

2nd best Student 2020/2021 academic year – Mfanstipim

PROFESSIONAL TRAINING AND CERTIFICATION

Build Chat Applications	365 Data science	ID: CC-EE76A0036
The Machine Learning Algorithms A-Z	365 Data science	ID: CC-A58D17499
Introduction to Deep Learning	DataCamp	https://tinyurl.com/4ax3zs7y
Time Series Forecasting	DataCamp	https://tinyurl.com/yc3ajkj7
Intermediate Deep learning	DataCamp	https://tinyurl.com/2mj6ewud

PROFESSIONAL AFFILIATIONS

- Society of Petroleum Engineers (SPE)-KNUST Chapter-2024/2025 Executive (Treasurer)
- Society of Petroleum Engineers- Active Member 2021 - present
- Young Oil Explorers – Member 2021-2023

VOLUNTARY ACTIVITIES

- **SPE Energyforme-** Visited basic schools to teach fundamental petroleum principles using simple household items for illustrations.
- **Tree planting Exercise** – Engaged in a school tree planting activity planting over 30 trees.

TECHNICAL SKILLS

- Programming Languages: *python, JavaScript, Html & CSS, Node.js, VBA*
- Advanced in Machine Learning and data science.
- Industry-Specific software: *AutoCAD, Prosper, Gap, Eclipse 100 &300.*
- Intermediate in deep learning.
- Financial Modelling with VBA in excel

SOFT SKILLS

- Creative problem solving through outside-the-box thinking.
- Excellent time management.
- Adaptability and willingness to learn new technologies and processes.
- Strong presentation abilities and active listening skills
- High attention to detail and excellent organizational skills.

HOBBIES

- Football
- Coding
- Reading

REFEREES

Dr. STEPHEN ADJEL, PhD

Principal Investigator, AI-DPRS

Lecturer, Department of Petroleum Engineering, KNUST

adjeistephen89@gmail.com

MR. FELIX ANDOH.

Tema Tank Farm (TTF) operator.

felix.amankwa-andoh@ttfghana.com